



Fig. 5: Phase-change memory chips from Micron in flat-pin and ball-grid array packages

memory cell can store five bits. This is not a new concept as the present-day Flash memories rely on an eight-charge level scheme to store three bits per memory cell.

In the real world

Micron Technology offers a number of PCM devices with both serial and parallel data interfaces. The P5Q serial interface devices use serial peripheral interface (SPI) for reading and writing the binary data. These are available in 32Mb, 64Mb and 128Mb versions in 16-pin surface-mount packages with operating speeds as high as 66 MHz. Although there are 16 pins in these packages, only eight pins are electrically active. The logical organisation of these memories is such that they can be used as slow but non-volatile DRAM memories or as Flash memories.

Replacing Flash memories with PCM will improve memory access times due to reduced latencies and increased bandwidths. System reliability

will also increase because Flash and other non-volatile memories usually require high programming voltages that can cause undesired electromagnetic interference effects.

It is very easy to use any of these devices in an embedded digital application as the standard SPI protocol, supported by most of the microprocessors and microcontrollers, is used for data storage and retrieval.

For more demanding, higher-speed applications, P8P parallel interface devices offer much higher-performance input/output (I/O) through parallel data transfer. These come in ball-grid array packages and are especially suitable for use in video card and server applications. Fig. 5 shows phase-change memory chips from Micron in both flat-pin and ball-grid array packages.

Thermal programming of PCM chips means that these devices cannot be pre-programmed before installing them on printed circuit boards as

reflow soldering process will raise the chip temperature above crystallisation temperature of the phase-change material. Thus, after reflow soldering, all memory cells will be found in the set state, holding binary '1'. This is even more certain with the new lead-free soldering process that operates at a higher temperature than the earlier one. This essentially means that PCMs have to be loaded with their content after board assembly, through an appropriate in-system programming interface.

Phase-change materials are widely used for re-writable non-volatile digital storage in optical disks. Re-writable CDs, DVDs and Blu-ray disks utilise a coating of phase-change material in which 0's and 1's are written thermally by a focused laser beam. The two states have different reflectivity and thus can be read by bouncing a laser beam off the recorded spots.

The electronic phase-change memories that take advantage of the difference in resistivity rather than reflectivity of the two phases of materials like GST are, however, only now beginning to appear, with Micron being the first company to offer parts that are in full production. As higher-capacity devices make their appearance, this memory technology is certain to replace NOR and NAND Flash memory in applications such as mobile phones, MP3 players, tablets and even computer servers. Technology analysts are forecasting that in the next two to four years, PCM chips and modules will drastically change the memory landscape, enabling lower-cost but higher-performance digital products of the future. ●

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